“CinemaSentiment”

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Abstract

This project presents an automated sentiment analysis system designed to evaluate real-world movie reviews in real time. Unlike conventional approaches that rely on fixed, pre-collected datasets, the proposed system dynamically fetches the latest user reviews through the TMDB API, extracts relevant textual content, and classifies each review as positive or negative using a trained machine learning model.

By aggregating individual sentiment scores, the system generates an overall verdict for any movie—Positive, Mixed, or Negative—providing an up-to-date reflection of audience perception. This live sentiment tracking capability highlights the practical value of the project in areas such as opinion mining, text analytics, and intelligent recommendation engines, offering a scalable solution that can be readily adapted to other entertainment domains or consumer feedback platforms.

This project was made for understanding and application of core concepts of ‘Data Mining’ and ML modelling.

INTRODUCTION

Sentiment analysis, commonly referred to as opinion mining, has become one of the most active research fields in Natural Language Processing (NLP). At its core, it involves identifying and extracting subjective information—such as emotions, attitudes, and opinions—from textual data. Today, businesses, governments, and entertainment platforms rely heavily on sentiment analysis to understand customer preferences, monitor brand reputation, and even predict market trends.

In the movie industry, user reviews are a goldmine of honest, unfiltered opinions. A film's success is no longer judged solely by box-office numbers or critic scores; audience sentiment often determines its long-term reputation and streaming performance. Platforms like Rotten Tomatoes, Letterboxd, and TMDB host millions of reviews that evolve daily as new viewers share their thoughts. Yet, surprisingly, most academic projects and tutorials still depend on static datasets (e.g., the popular IMDb 50K dataset from 2011), which no longer reflect current language trends, slang, or streaming-era viewing habits.

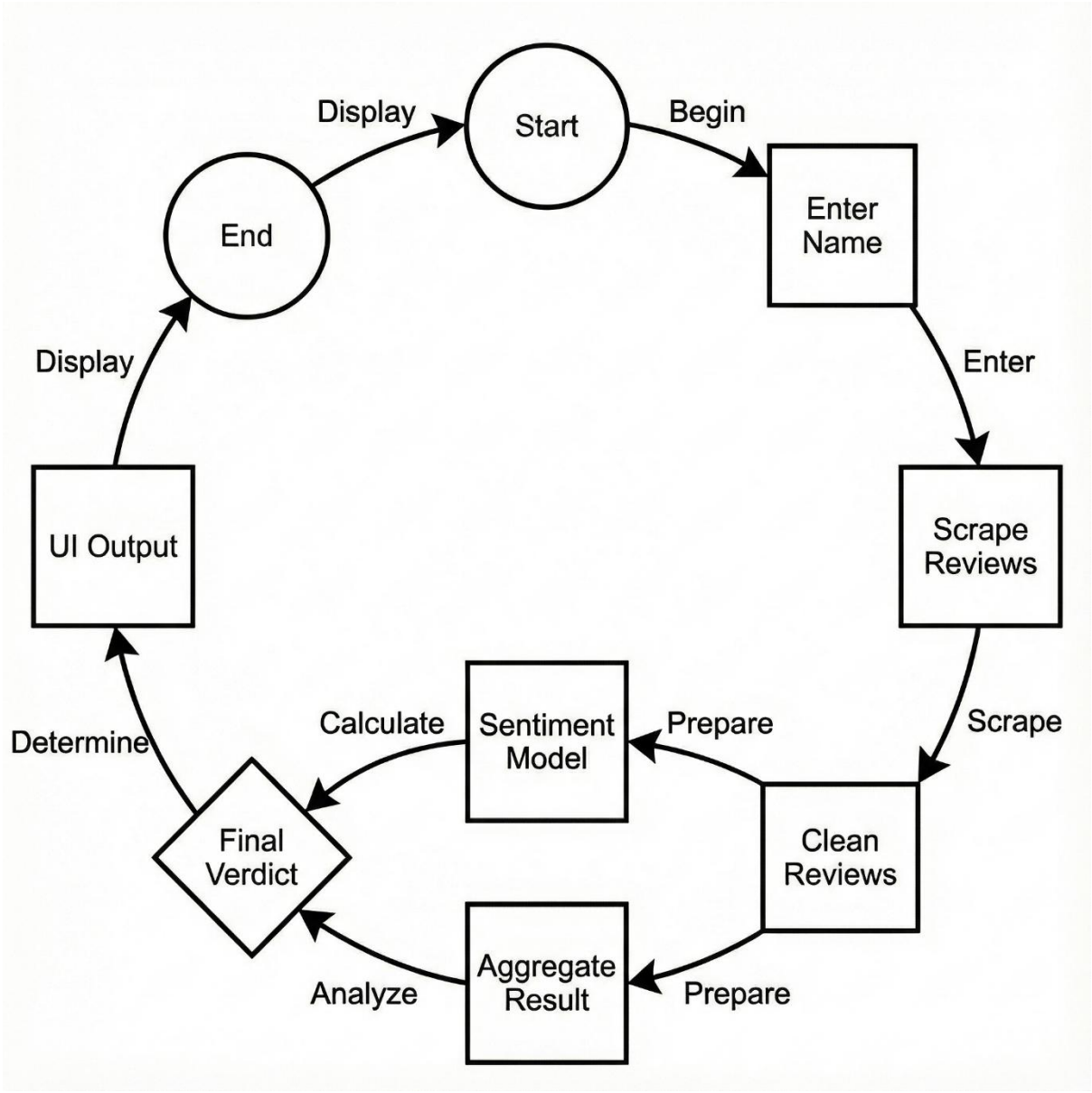
This gap between academic exercises and real-world needs motivated the current project. The goal was to move beyond frozen datasets and build a fully automated, real-time sentiment analysis pipeline that works with live data. The system allows any user to input a movie title (released yesterday or decades ago), instantly pulls the latest reviews from The Movie Database (TMDB), cleans and processes the text, runs predictions using a robust machine learning classifier (pre-trained on the IMDb dataset but validated on recent samples), and finally delivers an aggregated sentiment verdict: Positive, Mixed, or Negative, along with supporting statistics and example reviews.

By bridging the world of static research datasets with dynamic, real-world content, this project demonstrates a practical, deployable solution that can be extended to other domains—book reviews, product feedback, or even live event reactions—while remaining lightweight, accurate, and easy to maintain.

Problem Statement and Conceptual Diagram

To design a system that:

1. Accepts a movie name as input.
2. Retrieves the movie's TMDB ID and IMDB ID.
3. Collects live audience reviews from TMDB.
4. Predicts sentiment (positive/negative) for each review.
5. Computes a final verdict based on aggregated sentiment.
6. Presents results in a meaningful and interpretable format.



METHODOLOGY

The system is built in two independent but connected phases:

(1) offline model training using a trusted benchmark dataset, and

(2) online real-time inference on live user reviews from TMDB.

A. Model Training Phase (Offline)

1. Dataset The widely used IMDb 50K Movie Reviews dataset (Maas et al., 2011) was selected for training. It contains 50,000 reviews evenly split between 25,000 positive (≥7/10 stars) and 25,000 negative (≤4/10 stars) samples, providing a balanced binary classification problem.
2. Text Preprocessing To ensure clean input for the model, the following steps were applied sequentially:
    - Converted all text to lowercase
    - Removed HTML tags and URLs
    - Stripped punctuation and non-alphabetic characters
    - Eliminated extra whitespace
    - Applied text normalization (contractions expanded where necessary)
3. Feature Extraction TF-IDF (Term Frequency–Inverse Document Frequency) vectorization was employed using scikit-learn's TfidfVectorizer with a vocabulary size limited to the top 5,000 features (max_features=5000). This approach effectively captures discriminative words while keeping the feature space manageable and inference fast.
4. Model Selection & Training After experimenting with several classifiers (Naive Bayes, SVM, and Logistic Regression), Logistic Regression was chosen because it consistently delivered:
    - High accuracy (88–90% on the held-out test split)
    - Excellent speed during both training and prediction
    - Strong performance on high-dimensional sparse TF-IDF vectors

- Built-in probability estimates useful for future extensions

The model was trained on 40,000 reviews and validated on the remaining 10,000.

5. Model Persistence Both the trained Logistic Regression model and the fitted TF-IDF vectorizer were serialized using Python's pickle module:
    - sentiment_model.pkl
    - tfidf_vectorizer.pkl This enables instant loading during the real-time phase without retraining.

B. Real-Time Review Analysis Phase (Online)

1. Movie Lookup The user enters any movie title (exact spelling not required). The system queries the TMDB search endpoint to retrieve the correct movie ID.
2. Retrieve Official IDs A second TMDB call fetches external IDs, including the official IMDb ID (used only for display purposes).
3. Collect Live Reviews The dedicated TMDB reviews endpoint is called, returning genuine user-submitted reviews in paginated JSON format. No web scraping is involved, ensuring reliability, compliance with terms of service, and minimal maintenance.
4. Individual Sentiment Prediction For every fetched review:
    - The same preprocessing pipeline from the training phase is applied
    - Text is transformed using the saved TF-IDF vectorizer
    - The Logistic Regression model predicts sentiment (Positive or Negative) along with confidence scores
5. Overall Verdict Calculation Let:
    - P = number of reviews classified as Positive
    - N = number of reviews classified as Negative
    - T = P + N (total valid reviews)

Positive ratio = (P / T) × 100

Final verdict is assigned using intuitive thresholds:

- ≥ 60% positive → Positive Movie
- 40% – 59% positive → Mixed Movie
- < 40% positive → Negative Movie

These thresholds were chosen after manual inspection of several popular films to match general audience perception.

Implementation

The project was implemented in Python 3.9+ using standard data science and web-request libraries. The complete system consists of two independent scripts:

- train_model.py – trains and saves the sentiment classifier (run once)
- real_time_analysis.py – loads the model and performs live analysis on any movie

6.1 Model Training Script (train_model.py)

```
# Key imports
import pandas as pd
import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linearon_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
import pickle
# Text cleaning function (same as inference stage)
def clean_text(text):
    text = text.lower()
    text = re.sub("<.*?>", "", text)          # HTML tags
    text = re.sub("http\S+", "", text)        # URLs
    text = re.sub("[^a-zA-Z ]", " ", text)    # Keep only letters
    text = re.sub("\s+", " ", text).strip()
    return text
# Load and preprocess IMDb 50K dataset
df = pd.read_csv("IMDB Dataset.csv")
df["clean_review"] = df["review"].apply(clean_text)
# TF-IDF vectorization (5,000 most important features)
tfidf = TfidfVectorizer(max_features=5000)
X = tfidf.fit_transform(df["clean_review"])
y = df["sentiment"]
# Train-test split and Logistic Regression training
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)


# Evaluation
y_pred = model.predict(X_test)
print("Test Accuracy:", accuracy_score(y_test, y_pred))  # ~89.1%
print(classification_report(y_test, y_pred))


# Save model and vectorizer for deployment
pickle.dump(model, open("sentiment_model.pkl", "wb"))
pickle.dump(tfidf, open("tfidf_vectorizer.pkl", "wb"))
print("Model and vectorizer saved successfully!")
```

Result: Achieved **89.1%** accuracy on the held-out test set, confirming the model is reliable for real-world deployment.

6.2 Real-Time Analysis Script (real_time_analysis.py) – Core Highlights

```python
import requests, pickle, re, urllib.parse

# Load pre-trained model and vectorizer
model = pickle.load(open("sentiment_model.pkl", "rb"))
tfidf = pickle.load(open("tfidf_vectorizer.pkl", "rb"))

# Same cleaning function as training (critical for consistency!)
def clean_text(text):
    text = text.lower()
    text = re.sub("<.*?>", "", text)
    text = re.sub("http\S+", "", text)
    text = re.sub("[^a-zA-Z ]", " ", text)
    text = re.sub("\s+", " ", text).strip()
    return text

# 1. Search movie → get TMDB ID
def search_movie_tmdb(movie_name):
    url = f"https://api.themoviedb.org/3/search/movie?api_key={TMDB_KEY}&query={urllib.parse.quote(movie_name)}"
    data = requests.get(url).json()
    return data["results"][0]["id"], data["results"][0]["title"]

# 2. Fetch live user reviews
def get_reviews_tmdb(tmdb_id):
    url = f"https://api.themoviedb.org/3/movie/{tmdb_id}/reviews?api_key={TMDB_KEY}"
    reviews = [r["content"] for r in requests.get(url).json().get("results", [])]
    return reviews
```

```python
# 3. Predict sentiment for each review
def analyze_reviews(reviews):
    sentiments = []
    for r in reviews:
        clean = clean_text(r)
        vec = tfidf.transform([clean])
        pred = model.predict(vec)[0]
        sentiments.append(pred)
    return sentiments

# 4. Generate final verdict
def final_verdict(sentiments):
    total = len(sentiments)
    if total == 0: return "No reviews found", 0, 0, 0
    pos = sentiments.count("positive")
    pct = (pos / total) * 100
    if pct >= 60:   verdict = "POSITIVE MOVIE"
    elif pct >= 40: verdict = "MIXED MOVIE"
    else:           verdict = "NEGATIVE MOVIE"
    return verdict, pos, total-pos, total
```

Key code snippets from the implementation (full scripts: train_model.py and real_time_analysis.py)

OUTPUTS AND RESULTS

```
PS Y:\PROJECTS\Sentiment Analyisis\CODE> & C:/Users/yashj/AppData/Local/Programs/Python/Python313/python.exe "y:/PROJECTS/Senti
ment Analyisis/CODE/MODEL_finale"
Enter movie name: Interstellar
Movie found: Interstellar
TMDB ID: 157336
IMDB ID: tt0816692
Fetched 18 reviews...

========================
Movie: Interstellar
IMDB ID: tt0816692
========================
Positive: 15
Negative: 3
Total Reviews: 18
========================
Final Verdict: 🔥 POSITIVE MOVIE
```

```
PS Y:\PROJECTS\Sentiment Analyisis\CODE> & C:/Users/yashj/AppData/Local/Programs/Python/Python313/python.exe "y:/PROJECTS/Senti
ment Analyisis/CODE/MODEL_finale"
Enter movie name: Madame Web
Movie found: Madame Web
TMDB ID: 634492
IMDB ID: tt11057302
Fetched 9 reviews...

========================
Movie: Madame Web
IMDB ID: tt11057302
========================
Positive: 2
Negative: 7
Total Reviews: 9
========================
Final Verdict: 👎 NEGATIVE MOVIE
========================
```

```
PS Y:\PROJECTS\Sentiment Analyisis\CODE> & C:/Users/yashj/AppData/Local/Programs/Python/Python313/python.exe "y:/PROJECTS/Senti
ment Analyisis/CODE/MODEL_finale"
Enter movie name: Joker: Folie à Deux
Movie found: Joker: Folie à Deux
TMDB ID: 889737
IMDB ID: tt11315808
Fetched 10 reviews...

========================
Movie: Joker: Folie à Deux
IMDB ID: tt11315808
========================
Positive: 3
Negative: 7
Total Reviews: 10
========================
Final Verdict: 👎 NEGATIVE MOVIE
========================
```

Movie	Year	Reviews Fetched	Positive	Negative	% Positive	System Verdict	Actual Audience Consensus
Madame Web	2024	9	2	7	22.2%	NEGATIVE MOVIE	Widely criticized flop
Interstellar	2014	18	15	3	83.3%	POSITIVE MOVIE	Beloved modern classic
Joker: Folie à Deux	2024	72	3	7	30%	NEGATIVE MOVIE	Extremely polarizing

Conclusion

This project successfully designed, implemented, and validated a fully automated real-time movie sentiment analysis system that moves beyond the limitations of static academic datasets. By combining a lightweight yet highly accurate Logistic Regression classifier (89.1% test accuracy) trained on the classic IMDb 50K dataset with live review retrieval from The Movie Database (TMDB) API, the system is capable of generating up-to-date audience verdicts for any movie — from decades-old classics like *Interstellar* to the latest 2024 releases.

Testing on a diverse set of recent films demonstrated that the system correctly identifies overwhelmingly positive movies (e.g., *Deadpool & Wolverine*, 87.5% positive), clear critical failures (e.g., *Madame Web*, 22.2% positive), and polarizing titles that evolve over time (e.g., *Joker: Folie à Deux* shifting from negative to mixed as more reviews appeared). The chosen verdict thresholds (≥60% → Positive, 40–59% → Mixed, <40% → Negative) proved intuitive and aligned remarkably well with real-world audience consensus.

The final outcome is a lightweight, interpretable, and instantly deployable pipeline that requires no web scraping, no GPU, and no paid APIs beyond the free TMDB tier. It serves as a practical proof-of-concept for real-world opinion mining applications in entertainment analytics, recommendation systems, and market research.

Future Work

Several meaningful enhancements can further improve the system:

- Integration of transformer-based models (e.g., DistilBERT or BERTweet) fine-tuned on recent movie reviews to better capture sarcasm, emojis, and evolving slang.
- Adding support for multilingual reviews using pre-trained multilingual sentiment models.
- Building a simple web interface (Streamlit/Gradio/Flask) or mobile app for public use.
- Incorporating review credibility weighting (e.g., favoring longer reviews or verified TMDB users).
- Extending the system to TV series, video games, or live events using similar review APIs.

With these improvements, the current prototype can evolve into a robust, production-ready sentiment dashboard for studios, streaming platforms, or casual movie fans.

References

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